SLIDE 1 Title, Candidate Credentials & Examination Overview

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Good afternoon, members of the committee. *Brief pause*

My name is Jerome Dixon. I am a student in the Integrative Life Sciences Doctoral Program, conducting my dissertation research within the School of Pharmacy at Virginia Commonwealth University.

Welcome to my doctoral candidacy proposal defense: *Gradient-Boosted Trees and Consensus-Risk For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD*.

Across emergency medicine and outpatient pharmacotherapy, clinicians face an escalating dual crisis: the ongoing mortality of the opioid epidemic and the surging, silent morbidity of geriatric polypharmacy adverse drug events - ADEs. Today, I will present our computational framework addressing this challenge.

This research presents an end-to-end architecture that connects population-scale predictive analytics with explainable clinical decision support.

Using longitudinal statewide claims from the Virginia All-Payer Claims Database, the architecture integrates gradient-boosted tree ensembles across multiple topologies, rigorous temporal leakage audits, and a dual-model consensus attribution framework combining SHAP values with formal symbolic logic.

Lean in / Vocal emphasis

Our ultimate aim is to support actionable insights at the point of care while minimizing unnecessary alerts and clinician alert fatigue.

With that, let us proceed on our journey."

Transition *→ Directing your attention to Slide 2, let us examine how this research aligns across the five scientific disciplines represented on this committee.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Candidate: R. Jerome Dixon, MDA, MSc IT • **Chair:** Dr. Elvin T. Price • **Scope:** Nov 9 Candidacy (Aims 1–3, Design, Estimators) vs. Nov 23 Holdout Defense • **Deck:** `pgx.jerome-dixon.io/9NOV_presentation`

Untangled Interval & Inequality Strings: Verbalized Specific Aims 1–3, **IR**, and **IE** as natural phrases ("*Specific Aims one through three*", "*Intervention Rate and Interaction Effect*"). • **Streamlined Clinical Rationale:** Frame preventable emergency department utilization upfront before establishing the November 9 vs. November 23 scope firewall. • **Structured Funnel Pacing:** Deliver the three opening pillars (**First**, **Second**, **Third**) with deliberate vocal emphasis on "*predictive, data-driven pharmacovigilance*".

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Price) & Scope Firewall Pivot: "*While Chapters 3, 4, and 5 are published in Clinical and Translational Science, today's November 9 Candidacy Examination evaluates pre-specified Specific Aims (Aims 1–3), study design, mathematical estimators, and 4-pillar governance architecture. Complete empirical 2019 holdout tables, feature-level IR/IE rankings, and sensitivity findings are staged for our November 23 Final Dissertation Defense.*"

Scope Boundary Anchor: "*I use statewide Virginia APCD claims to develop and temporally evaluate CatBoost, XGBoost, and XGBoost-RF models across two distinct ED targets—opioid and geriatric polypharmacy events—identifying reproducible Consensus-Risk features via TreeSHAP intersection and Formal Feature Attribution rule support to power a zero-PHI medication-review prioritization prototype.*"

SLIDE 2 Supervisory Committee & Multidisciplinary Alignments

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Before we dive into the core modules, I want to express my sincere gratitude to my supervisory committee for your time, example, and invaluable mentorship throughout this journey. *Brief pause*

Your combined expertise across pharmacotherapy, systems engineering, informatics, and real-world evidence is vital to this work.

Lean in / Vocal emphasis

I'd like to specifically thank my chair and sponsor, Dr. Price, as well as Professor Bost, Dr. Gal, Dr. Wijesinghe, and Dr. DeShazo. Thank you all for being here today to evaluate this candidacy proposal."

Transition *→ With our academic alignment established, Slide 3 outlines the five core scientific modules that structure today's presentation.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Panel Mapping: Dr. Price (PGx, CPIC Drug Counts) • **Prof. Bost** (CRISP-DM, **O(T·L)** CNF) • **Dr. Gal** (DuckDB, Step 3b $\tau_{post} \geq 0.80$) • **Dr. Wijesinghe** (Multi-Drug Synergies) • **Dr. DeShazo** (APCD, TRIPOD+AI, CDS)

Untangled Interval & Inequality Strings: Verbalized **O(T · L)** as "*polynomial time proportional to tree count times leaf count*" so symbolic complexity is effortless to hear. • **Streamlined Clinical Rationale:** Connect each faculty member's methodology directly to the clinical pharmacovigilance pipeline before moving to the agenda. • **Structured Funnel Pacing:** Group the five faculty alignments into three clear signposted beats (**First**, **Second**, **Third**).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Cross-Disciplinary Routing: Route formal logic, **O(T · L)** CNF compilation, and Hessian loss ($h_i = p_i(1-p_i)$) to **Prof. Bost & Dr. Gal**; route non-causal MTM prioritization, Step 5 CPIC drug counts (`pgx_num_cpic_drugs`), and multi-drug synergy to **Dr. Price & Dr. Wijesinghe**; route statewide outcome capture and adjudicated pharmacy fills to **Dr. DeShazo**.

Committee Q&A Pivot (Bridging Pharmacy & Computer Science): "*Classical clinical CDS relies on expert pairwise rules that trigger 90%–95% alert fatigue overrides, whereas pure ML models operate as black boxes. Bridging both lets us use GBDT ensembles for dynamic risk discovery while compiling tree structures into verifiable Boolean Horn rules for clinical pharmacist review.*"

SLIDE 3 Roadmap

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"As shown on Slide 3, today's examination is structured into five sequential scientific modules that trace our end-to-end research lifecycle. *Brief pause*

First, in Modules I and II, we establish the dual clinical crises of acute opioid toxicity and geriatric polypharmacy—presenting our systematic review of 94 studies that uncovered 'The Empty Intersection'—and lock our three-tier governance framework uniting CRISP-DM, TRIPOD+AI, and the Clinical OODA Loop on statewide Virginia APCD claims.

Second, in Module III, we detail our 39-column medical and pharmacy table join, contrast our 12-month chronic opioid trajectory against our 21-day proximal polypharmacy funnel, and enforce our Step 3b temporal leakage audit.

Third, in Modules IV and V, we unpack our tri-model gradient-boosted tree geometry, claims-density stratification, and Consensus-Risk engine uniting dual-model TreeSHAP with symbolic Formal Feature Attribution, culminating in our zero-PHI serverless clinical decision support prototype, six-chapter published portfolio, and four methodological defense fences.

Lean in / Vocal emphasis

Each module builds directly upon the last to guarantee temporal validity, interpretability, and clinical actionability."

Transition *→ Let us enter Module I on Slide 4 and examine the clinical and literature gaps driving this research.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Roadmap (5 Core Modules): I. Problem • II. Governance & Data • III. Cohorts • IV. Modeling • V. Translation and Deployment

Untangled Interval & Inequality Strings: Expressed SHAP ∩ FFA and **n=94** verbally as "*uniting dual-model TreeSHAP with symbolic Formal Feature Attribution*" and "*94 studies*". • **Streamlined Clinical Rationale:** Present the 5 modules as a cohesive progression from clinical gap to bedside translation. • **Structured Funnel Pacing:** Signpost the roadmap in three clear macro-stages (**First**: Modules I–II, **Second**: Module III, **Third**: Modules IV–V).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. DeShazo) & Agenda Control Pivot: "*That is a fundamental question regarding our Aim 2 symbolic engine, Dr. Name. I have dedicated Slide 13 in Module IV specifically to the mathematical formulation of our PySAT Hitman solver and CNF tree compilation, and will address the exact mechanics when we reach that module.*"

Central Scientific Contribution Anchor: "*A methodological and applied workflow contribution: independent out-of-time temporal holdout validation, cohort-specific claims feature engineering, and an interpretable SHAP ∩ FFA consensus engine across two distinct medication-safety targets—not causal drug-interaction claims or an autonomous prescribing score.*"

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 1–3)

☐ Slide 1 ☐ Slide 2 ☐ Slide 3 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 1 of 3 (Front / Recto)

SLIDE 4 Module I — Upstream Clinical Crisis, Literature Gap & "The Empty Intersection"

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Turning to Slide 4, let us examine the clinical imperative and systematic literature gaps motivating this dissertation. *Brief pause*

In the United States, the opioid epidemic has claimed over 500,000 lives since 1999 at an annual economic cost exceeding 78.5 billion dollars, while geriatric polypharmacy—taking five or more concurrent medications—affects over 40% of older adults, driving 700,000 emergency visits and 125,000 deaths each year. Yet our systematic review of 94 supervised machine learning studies uncovered a severe structural imbalance: 95% focus on opioid use disorder diagnosis (46%) or general overdose utilization (49%), while only 5%—just five studies—address polypharmacy adverse drug events, and only 6 of 1,473 opioid records evaluate claims-based emergency prediction.

Furthermore, our PROBAST appraisal of 73 predictive models revealed three systematic failures:

First, 30% suffer from temporal data leakage—ingesting post-index treatments that artificially inflate AUROC by 5 to 28 percentage points.

Second, 81% omit pharmacogenomic modifiers such as *CYP2D6* and *OPRM1*.

Lean in / Vocal emphasis

Third, across all 73 models, exactly zero combine structured pharmacogenomic features, explainable AI, and out-of-time temporal holdout validation. Filling this 'Empty Intersection' is the central mission of our work."

Transition → *Slide 5 translates these literature gaps into two distinct clinical disease formulations and three specific aims.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

SQLR (n=94 Studies, n=73 PROBAST): 46% OUD Diagnosis, 49% Acute Overdose, **only 5% (n=5) Polypharmacy ADEs • 3 Failures:** 30% Temporal Leakage (+5–28 AUROC pts), 81% Omit PGx, **0/73 combine PGx + XAI + Temporal Holdout Validation**

Untangled Interval & Inequality Strings: Verbalized ≥ 5 drugs, ≥ 65, and n=0/73 naturally as *"five or more concurrent medications"*, *"older adults"*, and *"across all 73 models, exactly zero"*. • **Streamlined Clinical Rationale:** Establish the national mortality and cost burden first to make the 5% polypharmacy literature neglect immediately striking. • **Structured Funnel Pacing:** Deliver the three PROBAST failures with crisp signposting (**First**, **Second**, **Third**), leaning in on *"Empty Intersection"*.

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Price) — Literature Gap Pivot: *"Across 94 supervised ML studies in our Chapter 1 systematic review, 95% focus on OUD diagnosis (46%) or acute utilization/overdose (49%), while only 5% (**n=5**) address polypharmacy ADEs and only 6 of 1,473 OUD records evaluate claims-based ML for opioid ED visits. Across 73 PROBAST-assessed models, 30% suffer from temporal data leakage, 81% omit PGx modifiers, and 0 of 73 models ('The Empty Intersection') unite structured PGx, explainable AI, and out-of-time temporal holdout validation."*

Committee Q&A Pivot (Pervasive Temporal Leakage): *"Unscreened EHR and claims tables pool index encounter billing with pre-index predictors; without explicit event partitioning ($t_{claim} < t_0$), models latch onto point-of-care rescue drugs (naloxone, buprenorphine) and ED toxicology orders."*

SLIDE 5 Module I — Clinical Problem Formulation: Dual Crises, "Silent ADEs" & Specific Aims

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 5 outlines our clinical problem formulation and specific aims. *Brief pause*

To prevent clinical masking between substance-use disorders and geriatric pharmacotherapy, our study design separates analysis into two mutually exclusive Virginia APCD cohorts.

First, Cohort 1 (`opioid_ed` , ages 13 to 64) models chronic opioid risk over a 12-month trajectory driven by mu-opioid receptor tolerance, cumulative morphine milligram equivalent escalation, and explicit ICD-10 billing anchors— `F11.20` and `T40.0X` through `T40.4X` —inspected across all ten diagnostic positions.

Second, Cohort 2 (`ed_non_opioid` , ages 65 to 114) targets acute geriatric polypharmacy over a 21-day proximal window. Here we confront the 'Silent ADE' coding reality: fewer than 5% of drug-induced emergency visits in older adults carry explicit `T36` through `T50` poisoning codes. Instead, hospitals bill the downstream clinical syndrome—unexplained falls (`R29.6`), hip fractures (`S72`), syncope (`R55`), delirium (`F05`), gastrointestinal bleeding (`K92.2`), or acute kidney injury (`N17`)—which we anchor via Milliman HCG emergency lines `P51b` , `O11` , and `P33` with zero opioid codes.

Lean in / Vocal emphasis

Third, our three Specific Aims evaluate tri-model GBDT discrimination across both cohorts in Aim 1, extract Consensus-Risk features via TreeSHAP and symbolic rules in Aim 2, and deploy a serverless pharmacist triage prototype in Aim 3."

Transition → *Having defined our two clinical problems, Slide 6 enters Module II to establish the three-tier governance architecture governing our pipeline.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Dual Targets: `opioid_ed` (Ages 13–64, 12-mo lookback, F11.20 / T40.0X–T40.4X across 10 pos) vs. `ed_non_opioid` (Ages 65–114, ≥5 Rx, 21-d window, Milliman P51b/O11/P33, Silent ADEs: R29.6 falls, K92.2 GI bleed, N17 AKI; 0 opioid ICDs)

Untangled Interval & Inequality Strings: Spoke `ed_non_opioid` and `opioid_ed` clearly, and verbalized μ-opioid and <5% as *"mu-opioid"* and *"fewer than 5%"*. • **Streamlined Clinical Rationale:** Emphasize the distinct physiological and billing mechanisms (explicit overdose codes vs. downstream 'Silent ADE' syndromes) prior to stating the three aims. • **Structured Funnel Pacing:** Use **First** (`opioid_ed`), **Second** (`ed_non_opioid` & Silent ADEs), and **Third** (Specific Aims 1–3) to contrast the parallel cohorts cleanly.

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Wijesinghe & Dr. Price) — Dual Cohort Separation Rationale: *"Acute substance overdose and geriatric polypharmacy operate on incompatible time horizons (12-month chronic tolerance vs. 21-day acute metabolic saturation) and distinct billing mechanics (explicit F11/T40 codes vs. downstream 'Silent ADE' syndromes). Pooling them into a single multinomial model would distort feature attribution and impair clinical decision support."*

Committee Q&A Pivot (Capturing Silent ADEs Without T-Codes): *"By linking Milliman HCG emergency encounter lines (`P51b` , `O11` , `P33`) to active pre-index pharmacy dispenses within the 1–21 day FDA FAERS pharmacokinetic onset window while enforcing a 100% opioid code exclusion."*

SLIDE 6 Module II — Governance and Frameworks

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Directing your attention to Slide 6, our engineering and validation pipeline is governed by three concentric architectural rings. *Brief pause*

First, the outer ring executes the six-phase CRISP-DM systems engineering lifecycle, enforcing strict reproducibility across our ten-step codebase—from clinical problem formulation and DuckDB ingestion through Step 3b leakage filtering, tri-model GBDT fitting, and containerized deployment.

Second, the middle ring enforces the 27-item TRIPOD+AI biomedical reporting standard for Type 2b and Type 3 independent temporal holdout validation. Models are trained and tuned exclusively on 2016 through 2018 Virginia APCD claims and evaluated once on an untouched 2019 calendar holdout. Crucially, we prohibit post-hoc recalibration on the 2019 holdout: adjusting probability curves after inspecting test-period outcomes is equivalent to grading your own exam after peeking at the answer key, which masks temporal calibration drift.

Lean in / Vocal emphasis

Third, the inner core operationalizes the Clinical OODA Loop—Observe, Orient, Decide, and Act. Our GBDT models and symbolic Horn rules execute Observe and Orient, while licensed clinical pharmacists retain 100% authority over Decide and Act within an asynchronous medication review queue."

Transition → *Within this governance structure, Slide 7 justifies why Virginia's All-Payer Claims Database was selected over single-system hospital EHRs.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Governance and Frameworks: Outer: 6-Phase CRISP-DM (Steps 1a–10) • **Middle:** TRIPOD+AI Type 2b/3 (2016–2018 Train → untouched 2019 Holdout; 0 recalibration) • **Inner:** Clinical OODA Loop (100% Clinician Decide/Act)

Untangled Interval & Inequality Strings: Expressed temporal split bounds (`2016–2018` vs. `2019`) and validation tiers clearly without mathematical shorthand. • **Streamlined Clinical Rationale:** Use the exam-grading analogy to explain why prohibiting post-hoc recalibration on the 2019 holdout is essential for prospective clinical trust. • **Structured Funnel Pacing:** Move concentrically inward using **First** (Outer Ring: CRISP-DM), **Second** (Middle Ring: TRIPOD+AI), and **Third** (Inner Core: Clinical OODA Loop).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Prof. Bost) — 1-to-1 CRISP-DM Mapping: *"Phase 1 locks dual clinical targets under an OODA boundary; Phase 2 ingests statewide APCD claims and 'Silent ADE' syndromes; Phase 3 executes Step 3b leakage filtering and engineers CPIC drug counts; Phase 4 fits CatBoost, XGBoost, and XGBoost-RF geometries; Phase 5 executes TRIPOD+AI 2019 temporal validation and SHAP ∩ FFA consensus; and Phase 6 deploys our stateless AWS Lambda microservice."*

Committee Q&A Pivot (Zero Post-Hoc Recalibration Defense): *"In real-world prospective deployment, future outcomes are unobserved at prediction time; evaluating unadjusted Platt-scaled probabilities on 2019 tests true prospective model shelf-life under secular prescribing drift."*

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 4–6)

☐ Slide 4 ☐ Slide 5 ☐ Slide 6 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 1 of 3 (Back / Verso)

SLIDE 7 Module II — Why Virginia APCD Over EHR

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Turning to Slide 7, we justify selecting the Virginia All-Payer Claims Database over Electronic Health Records or single-payer Medicare datasets. *Brief pause*

Single-system EHRs suffer from critical outcome blindness in emergency prediction: when an individual overdoses or an older adult suffers a syncopal fall, emergency medical services transport them to the nearest hospital, not necessarily their home academic center.

First, the Virginia APCD provides a multi-payer, statewide window encompassing commercial, Medicare Advantage, and Medicaid populations—eliminating out-of-network emergency blindness across competing health systems with zero linkage attrition via Virginia Health Information.

Second, whereas EHRs record written e-prescriptions—missing the roughly 30% primary non-adherence drop-off abandoned at the pharmacy counter—APCD pharmacy claims capture paid, adjudicated dispenses that verify physical drug possession.

Lean in / Vocal emphasis

Third, while administrative claims lack raw DNA swabs and bedside vitals, we methodologically compensate in Step 5 (`5_pgx_analysis`) by mapping all dispensed APCD medications against the official CPIC directory (`drug_cpic_mapping_global.csv`) to determine whether each drug carries a CPIC guideline, computing patient-level counts of total drugs (`pgx_num_drugs`), CPIC-guideline drugs (`pgx_num_cpic_drugs`), and how many patients carry CPIC drugs—quantifying observable medication exposure burden directly without unscientific genotype imputation."

Transition *→ Slide 8 enters Module III to show how raw APCD Gold medical and pharmacy tables are relationally joined into our 39-column unified event fact table and pre-index feature schema.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Why Virginia APCD Over EHR: Statewide cross-system ED capture (0% out-of-network blindness; commercial, Medicare Adv. Medicaid) • Paid/adjudicated pharmacy fills (verifies possession vs. ~30% unfilled e-Rx) • Step 5 `pgx_num_cpic_drugs` compensation

Untangled Interval & Inequality Strings: Expressed linkage attrition and primary non-adherence percentages in clear spoken prose. • **Streamlined Clinical Rationale:** Contrast ambulance routing reality and pharmacy counter abandonment against statewide multi-payer claims capture before explaining the Step 5 CPIC compensation. • **Structured Funnel Pacing:** Use **First** (Multi-Payer Statewide Capture), **Second** (Adjudicated Physical Possession), and **Third** (Step 5 CPIC Drug Mapping Compensation).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Price & Dr. DeShazo) — Missing Lab/Genotype & Multi-Payer Pivot: *"The Virginia APCD provides a statewide multi-payer window across commercial, Medicare Advantage, and Medicaid populations that prevents single-system EHR selection bias and 30% unfilled prescription noise. In Step 5 (`5_pgx_analysis`), we map dispensed APCD drugs to whether or not they carry a CPIC guideline (`drug_cpic_mapping_global.csv`) and compute patient-level counts (`pgx_num_drugs` , `pgx_num_cpic_drugs`) and cohort counts of patients with CPIC drugs, capturing observable claims exposure directly without unscientific genotype imputation."*

Scope Boundary Anchor: *"Administrative race is never used as a proxy for genetic ancestry; verified CPIC drug counts test whether CPIC-guideline medication exposure predicts emergency risk, while patient-level allele cards are reserved for the November 23 final defense."*

SLIDE 8 Module III — Medical & Pharmacy Claims Schema

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Before examining our two target cohorts on Slides 9 and 10, Slide 8 details how we validate, clean, and relationally join the Virginia APCD Gold medical and pharmacy tables into our 39-column unified event fact table. *Brief pause*

To ensure administrative claims integrity prior to modeling, automated preprocessing scripts filter voided or reversed claim lines and harmonize schema fields across years.

First, the Gold medical table contributes 25 standardized columns—capturing all ten ICD-10 diagnostic positions, nine ICD procedure and three CPT modifier fields, Milliman HCG emergency service lines `P51b` , `O11` , and `P33` , and imputed demographics—while the Gold pharmacy table captures paid prescription fills normalized from National Drug Codes to generic RxNorm `drug_name` and `therapeutic_class_1` .

Second, in Phase 2, DuckDB links and unions these medical and pharmacy records at the patient level on `mi_person_key` , `event_year` , and `age_band` , backfilling demographic covariates across pharmacy rows and computing a global chronological `event_sequence` ordered by claim date relative to index date zero.

Lean in / Vocal emphasis

Third, from this strictly pre-index timeline—where claim service date precedes index date—we engineer our four core feature families: multi-position diagnosis and procedure indicators (`item_icd_` , `item_cpt_`), active generic drug exposures (`item_drug_*`), our Step 5 CPIC drug counts (`pgx_num_drugs` , `pgx_num_cpic_drugs`), and claims-density utilization tiers."

Transition *→ From this 39-column unified event fact table, Slide 9 derives our 12-month chronic opioid ED cohort (`opioid_ed`).*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Medical & Pharmacy Claims Schema (39-Col): Gold `medical` (25 cols: 10 ICD-10 Dx, 9 ICD-Proc, 3 CPT, HCG P51b/O11/P33) + Gold `pharmacy` (paid fills, generic `drug name`) joined on `(mi person key, event_year, age_band)` • Pre-index ($t < t_0$): `item_icd_*` , `item_cpt_*` , `item_drug_*` , CPIC counts, `n_events` strata

Untangled Interval & Inequality Strings: Expressed multi-year relational schema joins and ($t < t_0$) verbally as *"where claim service date precedes index date"* without reciting raw SQL window syntax. • **Streamlined Clinical Rationale:** Highlight claims deduplication, voided-claim filtering, and 10-position diagnostic capture as prerequisites for downstream predictive validity. • **Structured Funnel Pacing:** Use **First** (Gold Medical & Pharmacy Inputs), **Second** (39-Column Relational Join & Global Sequence), and **Third** (Pre-Index Feature Families).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Gal & Dr. DeShazo) — Schema, Claims Cleaning & Join Integrity: *"Automated preprocessing filtered voided/reversed claims and required continuous enrollment eligibility. By executing `UNION ALL` and person-level demographic backfill on `mi_person_key` prior to computing `event_sequence` , medical diagnoses, CPT procedures, and paid pharmacy dispenses share a single deterministic timeline relative to t_0."*

Committee Q&A Pivot (Retaining All 10 ICD-10 Diagnosis Positions): *"Primary billing positions frequently reflect hospital reimbursement priority rather than underlying drug toxicity; inspecting all ten diagnostic and nine procedure positions captures secondary overdoses, chronic pain, and antecedent adverse syndromes."*

SLIDE 9 Module III — Cohort Pipeline Architecture: Opioid-Related ED (opioid_ed 12-Month Model)

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 9 presents the architectural pipeline for Cohort 1, our 12-month chronic opioid emergency cohort, `opioid_ed` . *Brief pause*

To capture true acute opioid toxicity and dependence crises across adolescents and adults aged 13 to 64, we anchor index encounters using validated ICD-10 diagnostic criteria.

First, case inclusion requires an emergency claim carrying explicit ICD-10 `F11.20` opioid dependence or `T40.0X` through `T40.4X` acute opioid poisoning codes inspected across all ten diagnostic positions.

Second, because opioid tolerance and dose escalation evolve longitudinally, we extract a 12-month pre-index lookback window—from day minus 365 through day minus 1—paired with 1-to-5 incidence-density matched controls across eight age bands, sex, and calendar year, and screened via our Step 3b leakage purge.

Lean in / Vocal emphasis

Third, to process 1.8 Terabytes and 380 million claims without memory exhaustion, we engineered a partition-first out-of-core DuckDB pipeline on AWS Spot EC2 `x2iedn.8xlarge` workers. Partitioning by Age Band and Event Year eliminated shuffle skew—achieving a 15.1-fold speedup to 18.1 million records per minute, zero failures across 189 passes, and reducing compute cost from 23,000 dollars a month on EMR clusters to under 10 dollars per pass."

Transition *→ In Slide 10, we contrast this 12-month chronic pipeline with our 21-day proximal acute geriatric polypharmacy pipeline (`ed_non_opioid`).*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

`opioid_ed` Pipeline (12-Mo Model): APCD 1.8 TB Parquet (380M claims) on Spot EC2 `x2iedn.8xlarge` + DuckDB (15.1× speedup: 18.1M rec/min, 0/189 failures, <$10/pass) • t_0 Anchor (F11.20, T40.0X–T40.4X) • [t_0-365, t_0-1] + 1:5 Match (N=207,350) • Step 3b Purge → `safe_feature_filter.json`

Untangled Interval & Inequality Strings: Converted [t_0-365, t_0-1] to *"from day minus 365 through day minus 1"* and spoke $6363 × 3 = 189 passes and dollar amounts naturally. • **Streamlined Clinical Rationale:** Clarify why inspecting all ten diagnostic positions over a 365-day lookback captures chronic opioid escalation before presenting the DuckDB engineering benchmarks. • **Structured Funnel Pacing:** Use **First** (10-Position ICD-10 Case Anchor), **Second** (12-Month Window & 1:5 Matching), and **Third** (Out-of-Core DuckDB Speedup).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Gal) — Out-of-Core DuckDB Benchmarks: *"Partitioning by Age Band × Event Year on persistent Spot EC2 `x2iedn.8xlarge` workers eliminated Spark-style network shuffle skew, achieving 18.1M records/min (**15.1×** speedup), **<60s** Spot checkpoint recovery, and 0 partition failures across 189 execution passes at $6--10/\text{pass}$."*

Committee Q&A Pivot (Diagnostic Code Sensitivity & 12-Month Horizon): *"Relying on validated CDC surveillance codes (`F11.20` , `T40.0X – T40.4X`) across all 10 diagnostic positions captures overdoses presenting alongside secondary trauma, while the 12-month window captures progressive μ-opioid receptor tolerance and provider shopping."*

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 7–9)

☐ Slide 7 ☐ Slide 8 ☐ Slide 9 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 2 of 3 (Front / Recto)

SLIDE 10 Module III — Parallel Pipeline Architecture & 3-Stage Epidemiological Funnel (ed_non_opioid)

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Turning to Slide 10, we examine the parallel pipeline architecture for our geriatric polypharmacy cohort, `ed_non_opioid`.

Brief pause

Focusing on older adults aged 65 to 114, we anchor index encounters using Milliman HCG emergency service lines—specifically `P51b`, `O11`, and `P33`. To isolate pure medication-related adverse events from opioid toxicity, we apply an absolute opioid cleanse: excluding any patient with even a single opioid ICD-10 code across all ten diagnostic positions during the study year.

Next, we pass candidate records through a three-stage epidemiological funnel:

First, a high-utilization cap excluding individuals with seven or more annual emergency visits, filtering out chronic crisis utilizers.

Second, a 21-day proximal pharmacy window—from day minus 21 through day minus 1. This is empirically validated by our FDA FAERS and Virginia APCD time-to-onset convergence, establishing external-to-internal cohort validation for our polypharmacy pipeline—detailed in the Cohort Validation card and Figure 1B.

Lean in / Vocal emphasis

Third, we enforce a strict Zero-Day Discharge Guard where claim date precedes index date. This systematically purges any prescription filled on the calendar day of the emergency visit, guaranteeing that post-discharge medications are never mistaken for pre-visit etiology.

Brief pause

Cohort cases are matched 1-to-5 with controls across age band, sex, and calendar year. Finally, our verification script— `qa_index_date_uniqueness.py` —validates that every patient retains exactly one unique index date per calendar year."

Transition *→ With both cohorts derived and matched, Slide 11 enters Module IV to establish the mathematical geometry of our tri-model GBDT ensemble.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

`ed_non_opioid` Pipeline (21-Day Model): Ages 65–114 (≥5 Rx) → Milliman HCG (P51b, O11, P33) + **Fence 1:** 100% Opioid Cleanse (10 Dx pos) → **Fence 2:** <7 ED/yr cap + **Fence 3:** **[t_0-21, t_0-1]** window (>90% FAERS/APCD construct validation) + **Fence 4:** 0-Day Guard ($t_{claim} < t_0$) → 1:5 Match (**N=647,002**; `qa_index_date_uniqueness.py`)

Untangled Interval & Inequality Strings: Converted [t_0-21, t_0-1] to *"from day minus 21 through day minus 1"*, vocalized ($t_{claim} < t_0$) as *"where claim date precedes index date"*, and spoke ≥ 7 as *"seven or more annual emergency visits"*. • **Streamlined Clinical Rationale:** Clarified the rationale behind the Absolute Opioid Cleanse early in the passage (*"To isolate pure medication-related adverse events from opioid toxicity..."*) before detailing the 10-position criteria. • **Structured Funnel Pacing:** Delivered the 3-stage funnel with distinct verbal signposts (**First**, **Second**, **Third**) so the committee follows the sequential filtering steps without cognitive strain.

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Committee Q&A Pivot (Zero-Day Discharge Guard & `qa_index_date_uniqueness.py` Validation): *"Methodological Fence 4 (the Zero-Day Discharge Guard) strictly requires $t_{claim} < t_0$; any prescription filled on day t_0 is discarded because it represents discharge medications prescribed after the visit, not pre-visit etiology. Furthermore, `qa_index_date_uniqueness.py` programmatically verifies that no patient has duplicate index dates within a calendar year, preserving unit-of-analysis integrity."*

Committee Q&A Pivot (Milliman HCG `P51b` / `O11` / `P33`, Absolute Cleanse & 21-Day vs. 30/90-Day Window): *"Milliman HCG logic harmonizes emergency room categorization across commercial, Medicare Advantage, and Medicaid claims, eliminating hospital revenue-code (0450) billing artifacts. Full-year 10-position opioid cleansing prevents intermittent opioid confounding, and the FDA FAERS-supported 21-day window ([t_0-21, t_0-1]) prevents exposure misclassification from inactive 30/90-day historical fills."*

SLIDE 11 Module IV — Tri-Model GBDT Ensemble Geometry & Mathematical Formulation

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Turning to Slide 11, we present the mathematical formulation and tree geometries of our Step 6 tri-model framework. *Brief pause*

Unlike neural networks that construct continuous surfaces where formal rule extraction is computationally intractable, decision trees partition feature space into piecewise-constant hyperrectangles—a stepped staircase of decision regions that compiles directly into exact Boolean logic in polynomial time.

First, we evaluate three distinct tree architectures across 25-split stratified Monte Carlo cross-validation: CatBoost uses symmetric oblivious trees that apply the same split threshold across every node at a given depth to prevent overfitting on high-cardinality codes; XGBoost uses asymmetric depth-wise splits to isolate multi-drug synergies; and XGBoost-RF uses parallel bagging to reduce prediction variance in sparse utilization strata.

Second, our objective function optimizes binary logistic log-loss using second-order Taylor mechanics: the first derivative, or Gradient, measures residual error slope, while the second derivative, or Hessian, measures error surface curvature—acting like adaptive cruise control that takes bold steps on flat surfaces while braking near sharp curves.

Lean in / Vocal emphasis

Third, Step 6 ranks models strictly by Area Under the Precision-Recall Curve—PR-AUC—and Recall to lock the best CatBoost model and best XGBoost variant for consensus attribution."

Transition *→ Slide 12 details how our engineered features enter these trees and establishes the strict boundary isolating supervised predictors from exploratory visual analytics.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Tri-Model GBDT Equations: $f(x) = \sum w_i \mathbf{I}(x \text{ in } R_i)$ **(O(T·L)** CNF) • CatBoost (oblivious) vs. XGBoost (depth-wise) vs. XGBoost-RF (bagged) • Gradient $g_i = p_i - y_i$, Hessian $h_i = p_i(1-p_i)$, Newton Step $w_j^* = -\sum g_i / (\sum h_i + \lambda)$ • Selection: PR-AUC & Recall

Untangled Interval & Inequality Strings: Verbalized $f(x) = \sum w_i \mathbf{I}(x \text{ in } R_i)$, $g_i = p_i - y_i$, $h_i = p_i(1-p_i)$, and w_j^* as conceptual geometry and calculus analogies (*"stepped staircase of decision regions"*, *"residual error slope"*, *"error surface curvature"*) while formal equations stay on the slide. • **Streamlined Clinical Rationale:** Tie piecewise-constant tree geometry directly to polynomial-time clinical rule verification and multi-drug synergy capture. • **Structured Funnel Pacing:** Use **First** (Three Tree Geometries), **Second** (Second-Order Gradient/Hessian Mechanics), and **Third** (PR-AUC Selection under Class Imbalance).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Prof. Bost & Dr. Wijesinghe) — Why GBDTs over Deep Neural Networks or Logistic Regression? *"Multivariable logistic regression is the traditional epidemiological comparator but cannot capture higher-order 3- and 4-drug synergies without combinatorial manual terms, while deep neural networks build continuous manifolds where abductive rule extraction is co-NP-complete. Piecewise-constant GBDTs capture non-linear multi-drug synergies while permitting exact polynomial-time **$O(T \cdot L)$** CNF compilation."*

Committee Q&A Pivot (PR-AUC Prioritization & Class Imbalance vs. SMOTE): *"In imbalanced claims cohorts, AUROC is overly optimistic due to large true-negative denominators, whereas PR-AUC directly evaluates positive predictive value and sensitivity. Furthermore, we handle class imbalance via objective-function weighting (`scale_pos_weight`) and stratified CV rather than SMOTE, which distorts marginal distributions in sparse binary claims."*

SLIDE 12 Module IV — Feature Dynamics, Density Strata, Burstiness & Toolbox Demarcation

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 12 outlines our engineered feature dynamics and establishes an absolute methodological wall between supervised predictors and exploratory visual analytics. *Brief pause*

To ensure our models learn pharmacological risk rather than raw healthcare utilization volume, all supervised features are aggregated strictly prior to index date zero.

First, we transform raw pre-index event counts into four ordinal claims-density strata—Low, Medium, High, and Extreme, anchored to the 25th, 50th, and 95th training percentiles—and quantify prescribing cadence using our Inter-Claim Burstiness Index ranging from minus one for steady 30-day maintenance refills to plus one for sudden crisis clusters.

Second, we incorporate our Step 5 global drug-to-CPIC guideline mapping alongside patient-level counts of total drugs (`pgx_num_drugs`) and CPIC-guideline drugs (`pgx_num_cpic_drugs`).

Lean in / Vocal emphasis

Third, look at Part 2: exploratory visual tools— `bupaR` process mining, Dynamic Time Warping, and `FP-Growth` itemset mining—are strictly excluded from the supervised GBDT predictor matrix and never compute Step 8 Intervention Rate or Interaction Effect metrics. Because cross-patient sequence alignments and pooled itemset networks span across patients, fencing them exclusively into Step 3b quality assurance and Step 9 visual dashboards preserves strict row-wise sample independence."

Transition *→ Slide 13 presents the core theoretical innovation of Aim 2: our Consensus-Risk attribution engine and consensus risk attenuation (**IR**) and interaction (**IE**) estimators.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Feature Space vs. Exploratory Wall: Density bins $b(n_{ev})$ in {0,1,2,3} (P_{25}, P_{50}, P_{95}) • Burstiness $B = (\sigma_{\Delta t} - \mu_{\Delta t})/(\sigma_{\Delta t} + \mu_{\Delta t})$ in [-1,1] • Step 5 CPIC counts (`pgx_num_drugs`, `pgx_num_cpic_drugs`) • **Excluded:** bupaR, D_{DTW}, $Lift_{FPG}$

Untangled Interval & Inequality Strings: Verbalized $b(n_{ev})$ in {0,1,2,3} and $B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}}$ in [-1, 1] as *"four ordinal claims-density strata"* and *"ranging from minus one for steady 30-day maintenance refills to plus one for sudden crisis clusters"*. • **Streamlined Clinical Rationale:** Explain why raw utilization volume must be stratified before contrasting row-wise supervised features against cross-patient visual analytics. • **Structured Funnel Pacing:** Use **First** (Density Strata & Burstiness), **Second** (Step 5 CPIC Drug Counts), and **Third** (Strict Exploratory Visual Demarcation).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Gal & Prof. Bost) — Toolbox Demarcation Pivot: *"Supervised GBDT matrices require strict row-wise sample independence prior to t_0. Cross-patient sequence alignments (`DTW` D_{DTW}), Markov process graphs (`bupaR`), and unsupervised itemset networks (`FP-Growth` $Lift_{FPG}$) pool trajectories across patients; therefore, they are strictly fenced off for Step 3b QA and Step 9 dashboard visualizations, while Step 8 **IR** and **IE** are computed exclusively from tree and AXP counterfactuals."*

Scope Boundary Anchor: *"Ordinal density binning compresses extreme right-skewed utilization tails into four monotonic strata (`Low`, `Medium`, `High`, `Extreme`), preventing raw encounter volume from monopolizing top-level tree splits."*

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 10–12)

☐ Slide 10 ☐ Slide 11 ☐ Slide 12 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 2 of 3 (Back / Verso)

SLIDE 13 Module IV — Consensus-Risk Integration: Dual-Model SHAP, Symbolic FFA & IR/IE Estimators

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 13 details the core theoretical contribution of Aim 2: our Consensus-Risk attribution engine. *Brief pause*

Standard impurity metrics like Gini gain fail under collinearity, and single explainable AI algorithms can split credit arbitrarily across correlated prescriptions. We resolve this with a three-stage dual-key security lock.

First, in Step 7, we compute game-theoretic TreeSHAP attributions across both the winning CatBoost model and winning XGBoost variant, retaining only features that clear the 75th percentile in both architectures.

Second, in Step 8, we compile the root-to-leaf paths of the winning XGBoost model into Conjunctive Normal Form clauses and apply the PySAT `Hitman` minimal hitting-set solver to extract irreducible Abductive Explanations—or AXPs—requiring rule support of at least 5% and confidence of at least 70%, combined with equal 50-50 weighting in Step 10. Think of TreeSHAP as an economist measuring marginal risk shifts across coalitions, and Formal Feature Attribution as a structural engineer proving the exact Boolean circuit that trips the breaker.

Lean in / Vocal emphasis

Third, to address Specific Aim 2 strictly via in-model consensus risk sensitivity contrasts ($\Delta\bar{p}$) from tree and AXP perturbations—expressed as consensus risk shifts rather than causal treatment effects—Intervention Rate measures single-feature predicted risk shift when set to its baseline median, while Interaction Effect uses second-order finite differences and Class-1 AXP co-occurrence lift above one to quantify super-additive model synergy."

Transition *→ Slide 14 demonstrates how our Step 3b temporal event-partitioning audit protects this attribution engine from post-index target leakage.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Aim 2 Consensus-Risk (SHAP ∩ FFA): Dual-Model $\text{supp}(\text{SHAP}_{CB}) \cap \text{supp}(\text{SHAP}_{XGB}) \geq Q_{75}$ • PySAT `Hitman` O(T·L) CNF AXPs (**Support ≥ 0.05, Conf ≥ 0.70**) • Estimators: $IR_{\Delta p}$, IR_{AXP}, $\Delta_{11} \approx \partial^2 p / \partial f_i \partial f_j$, **IE > 1.0** • **Score = $0.5 \cdot \text{SHAP}_{norm} + 0.5 \cdot \text{FFA}_{norm}$**

Untangled Interval & Inequality Strings: Verbalized ≥ Q_{75}, Support ≥ 0.05, Conf ≥ 0.70, $\Delta_{11} \approx \frac{\partial^2 \bar{p}}{\partial f_i \partial f_j}$, and IE > 1.0 as *"clear the 75th percentile", "support of at least 5% and confidence of at least 70%"*, and *"second-order finite differences and Class-1 AXP co-occurrence lift above one"*. • **Streamlined Clinical Rationale:** Explain why Gini gain and single XAI methods fail under collinearity before introducing the Economist (TreeSHAP) + Structural Engineer (PySAT FFA) lock. • **Structured Funnel Pacing:** Use **First** (Step 7 Dual-Model TreeSHAP), **Second** (Step 8 PySAT Hitman AXPs & Step 10 Synthesis), and **Third** (Consensus Risk IR & IE Estimators).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Prof. Bost, Dr. Wijesinghe & Dr. Gal) — Consensus Threshold Dial Pivot: *"In Aim 2, I define the Consensus-Risk thresholds—mean |SHAP| ≥ Q_{75}, FFA rule support ≥ **0.05**, and rule confidence ≥ **0.70**—explicitly as an operational precision–volume dial rather than a fixed biological constant, allowing health systems to tune review volume to pharmacist capacity. We validate this baseline operating point via Two-Gate Mirror Ablation (**SHAP**-only vs. **FFA**-only vs. **SHAP ∩ FFA**) and a Utilization-Free Specification Stress Test."*

Committee Q&A Pivot (TreeSHAP vs. Gini Impurity & IR vs. IE Distinction): *"Standard Gini impurity favors high-cardinality features and handles collinearity inconsistently, whereas TreeSHAP provides additive Shapley attributions. **IR** measures single-feature consensus risk attenuation ($IR_{\Delta \bar{p}}$ and IR_{AXP} when $f_i \rightarrow x_{med}$), whereas **IE** measures multi-feature synergy via $\Delta_{11} \approx \frac{\partial^2 \bar{p}}{\partial f_i \partial f_j}$ and Class-1 AXP lift $IE(f_i, f_j) > 1.0$."*

SLIDE 14 Module III/IV — Target Leakage Defense: The Step 3b Event-Partitioning Protocol

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 14 details our Step 3b temporal leakage audit—the engineering firewall that eliminates reverse causality. *Brief pause*

Following an acute emergency overdose, patients receive immediate clinical rescue protocols—such as emergency naloxone administration, rapid buprenorphine induction, and point-of-care urine toxicology screens. In 30% of published studies, models inadvertently ingest these post-index treatments, inflating AUROC by 5 to 28 percentage points simply by detecting the emergency response after the patient arrives.

First, Step 3b partitions every candidate code's timestamps around index date zero and computes its post-target ratio: the proportion of claims occurring on or after the emergency visit.

Second, any feature with a post-target ratio of 80% or higher is systematically purged. Across 1,029 candidate features, Step 3b purged 348 post-index confounders—or 33.8%—including 164 medications such as Suboxone at 100% post-index, 84 CPT codes such as urine toxicology `80307` at 100% post-index, and 100 secondary diagnostic codes.

Lean in / Vocal emphasis

Third, the remaining 682 verified pre-index antecedents—where claim date strictly precedes index date—are locked in `safe_feature_filter.json` and applied symmetrically across cases and controls, cutting the movie frame strictly before the emergency room doors open."

Transition *→ In Slide 15, we enter Module V to translate our leakage-free GBDT models and symbolic Horn rules into a serverless clinical decision support microservice.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Step 3b Leakage Formula: $\tau_{post}(c) = \sum I(t_{claim} \geq t_0) / [\sum I(t_{claim} < t_0) + \sum I(t_{claim} \geq t_0)] \geq 0.80$ → **Purge** • Evaluated 1,029 → Purged 348 (33.8%: 164 Rx [Suboxone 100%], 84 CPT [80307 100%], 100 ICD) → Locked 682 in `safe_feature_filter.json`

Untangled Interval & Inequality Strings: Verbalized $\tau_{post}(c) \geq 0.80$ and $t_{claim} < t_0$ as *"post-target ratio of 80% or higher"* and *"where claim date strictly precedes index date"*. • **Streamlined Clinical Rationale:** Anchor the leakage defense in clinical emergency department workflow and the *"Movie Frame Cut"* analogy before reciting the 348/1,029 purge tally. • **Structured Funnel Pacing:** Use **First** (Timestamp Partitioning), **Second** (80% Cutoff & 348 Feature Purge), and **Third** (682 Whitelist Locked in `safe_feature_filter.json`).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Gal) — Exact Step 3b Audit Breakdown: *"Evaluating 1,029 candidate features with $\tau_{post}(c) \geq 0.80$, we purged 348 post-index confounders (33.8%): 164 post-index medications (e.g., Suboxone, 6,082 claims at 100% post-index), 84 point-of-care CPT procedures (e.g., CPT 80307 toxicology, 3,636 claims at 100% post-index), and 100 secondary ICD codes, locking 682 clean pre-index antecedents in `safe_feature_filter.json`."*

Symmetric Application Anchor: *"The antecedent whitelist (`safe_feature_filter.json`) is derived on development data and applied identically across all cases and 1:5 matched controls to prevent differential feature availability."*

SLIDE 15 Module V — Aim 3 Value-Driven Clinical Decision Architecture & Workflow Integration

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Slide 15 presents Specific Aim 3: translating our validated modeling and attribution insights into an actionable clinical decision-support prototype. *Brief pause*

Traditional EHR drug-interaction checkers fire synchronous pop-up interrupts during rapid clinical encounters, suffering 90% to 95% clinician override rates. Furthermore, because administrative claims carry billing latency, claims-based consensus models are designed to support proactive Medication Therapy Management rather than acute bedside emergency triage.

First, our interface operationalizes the Clinical OODA Loop—Observe, Orient, Decide, and Act—delivering information structured for rapid clinical review while preserving 100% human clinician authority.

Second, we implement a Lean value stream framework, ranking and laying out information from high to low and left to right by clinical value—pairing patient event-density tiers with verifiable Boolean Horn rules rather than unexplained probability scores.

Lean in / Vocal emphasis

Third, an interactive 'What-If' scenario exploration view displays model-predicted risk shifts when candidate regimen features are toggled, supporting pharmacist medication review and evaluating explanation fidelity and clinical usability rather than claiming clinical outcomes effectiveness."

Transition *→ Slide 16 presents our peer-reviewed 6-chapter dissertation portfolio and 7-notebook codebase traceability.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Aim 3 Value-Driven Decision Architecture: OODA information delivery • Lean value stream ranking (high to low, left to right by value) • Event density strata • Pairwise pop-up fatigue eradication via sound Boolean Horn rules

Untangled Interval & Inequality Strings: Verbalized **Refill Gap ≤ 14d** and $\Delta \bar{p}$ as *"refill gap of 14 days or less"* and *"model-predicted risk shifts"*. • **Streamlined Clinical Rationale:** Emphasize the OODA loop and Lean value stream delivery frameworks upfront before explaining the verifiable Boolean Horn rules. • **Structured Funnel Pacing:** Use **First** (Clinical OODA Delivery), **Second** (Lean Value Stream Layout & Horn Rules), and **Third** (Interactive What-If Scenario Exploration).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. DeShazo & Dr. Price) — Associational Scenario Delta ($\Delta \bar{p}$) & Claims Lag vs. Real-Time Triage Pivot: *"Because administrative claims exhibit adjudication latency, our consensus-risk engine is designed for proactive ambulatory MTM population management rather than acute ED bedside triage. Furthermore, the scenario delta ($\Delta \bar{p} = \bar{p}_{ens}(x) - \bar{p}_{ens}(x_{f_i \rightarrow med})$) represents an associational profile comparison within the fitted ensemble—not an identified causal treatment effect—preserving 100% human clinician authority."*

Sparsity Resilience Anchor: *"Pre-computed cohort-median imputation masks in S3 metadata maintain stable risk triage up to 70% feature sparsity."*

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 13–15)

☐ Slide 13 ☐ Slide 14 ☐ Slide 15 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 3 of 3 (Front / Recto)

SLIDE 16 Module V — Dissertation Progress: 6-Chapter Published Portfolio & Codebase Traceability

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"Turning to Slide 16, this dissertation is structured as a peer-reviewed six-chapter manuscript portfolio backed by an end-to-end traceable codebase. *Brief pause*

First, Chapters 3, 4, and 5—representing the direct execution of Specific Aims 1, 2, and 3—are peer-reviewed and published in *Clinical and Translational Science* (`cts.70690` , `cts.70718` , and `cts.70697`), while Chapter 1 is an invited systematic review at the *Journal of Personalized Medicine*, Chapter 2 is under review at the *Journal of Big Data*, and Chapter 6 synthesizes all five chapters.

Second, every analytical step is reproducible across seven structured Jupyter notebooks in our `pgx-analysis` repository—spanning cohort filtering, Step 3b leakage screening, tri-model training, sensitivity validation, exploratory visual analytics, and AWS Lambda deployment—persisted in S3 Parquet medallion storage.

Lean in / Vocal emphasis

Third, while Chapters 3, 4, and 5 have passed external peer review, I explicitly reaffirm for the record that this Supervisory Committee holds sovereign academic authority over this dissertation. *Brief pause* I welcome the committee's recommendations on any modifications, refinements, or additional analyses to incorporate into the dissertation."

Transition → *Finally, Slide 17 synthesizes our core epidemiological and methodological safeguards, reiterates our candidacy scope demarcation, and opens the floor for oral examination.*

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

6-Chapter Portfolio & Status: Ch 1: SQLR (*J Pers Med*, Invited) • **Ch 2:** Biq Data Lifecycle (*J Big Data*, In Review) • **Ch 3 (Aim 1):** *Clin Transl Sci* (`CTS.70690`) • **Ch 4 (Aim 2):** *CTS* (`CTS.70718`) • **Ch 5 (Aim 3):** *CTS* (`CTS.70697`) • **Ch 6:** Synthesis (Drafted)

Untangled Interval & Inequality Strings: Expressed manuscript DOIs and notebook sequence cleanly without reading raw URL strings aloud. • **Streamlined Clinical Rationale:** Present peer-reviewed publications as external validation of the methodology while immediately affirming the Supervisory Committee's sovereign authority. • **Structured Funnel Pacing:** Use **First** (6-Chapter Portfolio), **Second** (7-Notebook S3 Reproducibility), and **Third** (Committee Sovereign Authority).

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Targeted Faculty Callout (Dr. Price & Full Committee) — Committee Sovereignty Pivot: *"While Chapters 3, 4, and 5 are published peer-reviewed manuscripts in Clinical and Translational Science, peer-reviewed publication is a scientific milestone, not a final boundary. The Supervisory Committee retains sovereign authority over the unifying synthesis, sensitivity analyses, and final dissertation compilation in Chapter 6 for our November 23 defense."*

Reproducibility Anchor: *"Every notebook reads and writes immutable Zstandard-compressed Parquet checkpoints and JSON manifests (* `safe_feature_filter.json` , `best_xgb_variant` *) in* `s3://pgxdatalake/gold/` *."*

SLIDE 17 Module V — Study Boundaries & Defense Conclusion

60–90s Podium Script

SPOKEN PRESENTER SCRIPT (60–90 SECONDS — UNTANGLED VERBATIM)

"In conclusion, on Slide 17, our dissertation is bounded by eight interlocking safeguards—four epidemiological cohort constraints on the left, and four clinical governance principles on the right:

On the left, our epidemiological design enforces:

First, the Absolute Opioid Cleanse across all ten diagnostic positions in our geriatric cohort;

Second, High-Utilization Capping excluding patients with seven or more annual emergency visits;

Third, the 21-Day Proximal Pharmacy Window—where external FDA FAERS onset kinetics directly validate our cohort derivation logic; and

Fourth, the Zero-Day Discharge Guard systematically purging day-zero fills and post-index confounders.

Brief pause

On the right, our governance framework establishes:

Fifth, Aim 3 is decision-support triage evaluating explanation fidelity, not a patient outcomes trial;

Sixth, Consensus Attribution quantifies predictive consensus risk for clinical review rather than causal treatment effects;

Seventh, the untouched 2019 temporal holdout evaluates real-world model transportability with zero recalibration; and

Eighth, our pharmacogenomic features evaluate claims-derived CPIC medication exposure rather than inferred germline genotypes.

Lean in / Vocal emphasis

By grounding our cohort logic in convergent FAERS surveillance kinetics and enforcing strict clinical governance, this dissertation advances both health data science methodology and public health pharmacovigilance. I welcome the committee's questions and discussion."

SLIDE VISUAL ANCHORS & ORAL DELIVERY / VOCAL NOTES

Study Boundaries (8 Core Safeguards): (1) 100% Opioid Cleanse • **(2)** <7 ED Cap • **(3)** 21-Day Lookback (FAERS Validated) • **(4)** 0-Day Guard ($t_{claim} < t_0$) • **(5)** Aim 3 CDS (Not Trials) • **(6)** Attribution (Not Causation) • **(7)** 2019 Holdout (0 Recalibration) • **(8)** Claims CPIC (Not Genotypes)

Untangled Interval & Inequality Strings: Fully verbalized [t_0-21, t_0-1] as *"from day minus 21 through day minus 1"*, ($t_{claim} < t_0$) as *"where claim date precedes index date"*, and ≥ 7 visits as *"seven or more annual emergency visits"*. • **Streamlined Clinical Rationale:** Deliver the 4 Methodological Fences as the foundational pillars uniting internal epidemiological validity and external clinical governance. • **Structured Funnel Pacing:** Recite the four fences using clear signposting (**First**, **Second**, **Third**, **Fourth**), dropping cadence slightly for a formal, confident defense conclusion.

STRATEGIC DEFENSE TIPS & COMMITTEE Q&A PIVOTS

Final Defense Anchor (Epidemiological & Governance Fences): *"The core contribution of this dissertation is not merely applying gradient boosting to claims data, but demonstrating how strict, biologically grounded epidemiological fences (Absolute Cleanse, High-Utilization Cap, 21-Day FAERS Window, and Zero-Day / Step 3b Discharge Guard) prevent common data leakage and reverse causality traps in health data science."*

Master Pivot — Non-Causal Prioritization & Untouched 2019 Holdout: *"Observational claims identify antecedent signals for pharmacist MTM prioritization without claiming causal treatment effects, and our untouched 2019 holdout evaluates real-world model shelf-life with zero post-hoc recalibration."*

DEFENSE GUIDE Oral Examination Quick-Reference: Pacing Budget, Scope Firewall & Committee Routing

Nov 9 Candidacy vs. Nov 23 Defense

MASTER SCOPE FIREWALL & MODULE PACING SCHEDULE (22 CORE SLIDES / 30 SLIDES DECK = 22–25 MIN PACED DELIVERY)

1. Candidacy vs. Final Defense Firewall: Today's **November 9 Candidacy Examination** evaluates pre-specified Specific Aims (**Aims 1–3**), study design, mathematical estimators, cohort funnels, and 4-pillar governance architecture. Empirical 2019 holdout tables, feature-level IR/IE rankings, and post-execution sensitivity sweeps are staged for the **November 23 Final Dissertation Defense**.

2. Module Pacing Checkpoints: Orientation (Slides 1–3): 0:00–4:00 • **Module I Problem, Gaps & Aims (Slides 4–11):** 4:00–8:00 • **Module II Governance & APCD (Slides 12–13):** 8:00–10:30 • **Module III Cohorts & Schema (Slides 14–16):** 10:30–14:30 • **Module IV GBDT, PGx & Leakage (Slides 17–19):** 14:30–19:00 • **Module V CDS, Portfolio & Boundaries (Slides 20–22):** 19:00–23:00.

Oral Exam Floor Open → *Turn to Chair Dr. Elvin Price to initiate committee questioning rounds.*

COMMITTEE MEMBER QUESTION ROUTING CHEAT-SHEET

- **Dr. Elvin Price (Chair — PGx / Geriatrics):** Step 5 CPIC drug mapping & patient counts (`pgx_num_drugs` , `pgx_num_cpic_drugs`), Silent ADE syndromes (falls R29.6, GI bleed K92.2, AKI N17), Aim 3 Value-Driven Clinical Decision Architecture & CPIC exposure counts.
- **Dr. Tamas Gal (Biostatistics / Informatics):** Step 3b event partitioning ($\tau_{post}(c) \geq 0.80$ purge, DADR 2023), `safe_feature_filter.json` , 0-Day Guard ($t_{claim} < t_0$), DuckDB 18.1M rec/min.
- **Prof. Franklin Bost (Systems Eng / XAI):** Piecewise-constant trees, **O(T · L)** CNF compilation, PySAT Hitman minimal AXPs, AWS Lambda (<$5/mo, 6 ms warm).
- **Dr. Wijesinghe & Dr. DeShazo (AI Sim / RWE):** Burstiness **B in [-1,1]**, Δ_{11} synergy, statewide Virginia APCD multi-payer coverage vs. EHR blindness, adjudicated fills.

NOTES SECTION SPEAKER & COMMITTEE NOTES (Slides 16–17 + Defense Guide)

☐ Slide 16 ☐ Slide 17 ☐ Price ☐ Gal ☐ Bost ☐ Wijesinghe ☐ DeShazo

1. Delivery / Pacing Note:

2. Committee Question / Observation:

3. Follow-Up Action / Defense Revision:

DUPLEX & PACING CUES

- Pause & make eye contact at transition
- Defer 2019 holdout tables → Nov 23
- Sheet 3 of 3 (Back / Verso)